

Lab 6: A Spectrogram and a Tiny Fusion

STFT of a sine mixture, patch it, then concat versus add two modalities.

Work in a **Jupyter** notebook with **numpy**, **PyTorch**, and `matplotlib`. Do notes **6.1–6.3** first. No pretrained Whisper or CLAP: construct the waveform yourself.

First time you implement this. The lecture notes this week already answered what the method is, why it exists, the architecture, the formula, and the tradeoffs. Read that first-time block before these exercises. This lab is the first time you **compute** it, not the first definition.

```
import numpy as np
import torch
import matplotlib.pyplot as plt
```

1. A sine mixture

Sampling rate $f_s = 8000$, one second.

```
fs = 8000
t = np.arange(0, 1.0, 1 / fs)
x = 0.6 * np.sin(2 * np.pi * 440 * t) + 0.4 * np.sin(2 * np.pi * 880 * t)
```

1. Plot `x[:800]` (0.1 s). Guess where the beats sit before you zoom.
2. Write an STFT with a Hann window, `n_fft=256`, `hop=128`. For each frame i , take `np.fft.rfft` of the windowed slice. Stack **magnitudes** into `S` with shape `(n_freq, n_frames)`.
3. Plot `np.log1p(S)` as an image (`origin="lower"`, `aspect auto`). You should see two horizontal bands near 440 Hz and 880 Hz. Convert a frequency bin index to hertz with $f = k \cdot f_s / N$.

2. Patches as tokens

Treat `S` as a one-channel image.

1. Crop `S` so both axes are divisible by 8. Cut non-overlapping 8×8 tiles. How many tokens N ? Flatten each tile to length 64.
2. Map tiles with `nn.Linear(64, 16)` (random weights are fine). Print the tensor shape `(N, 16)`. This is the ViT move from Lab 4, on a spectrogram instead of RGB.

3. Concat versus add

Two modality vectors in 4-D (pretend audio and text):

```
a = torch.tensor([1.0, 0.0, 1.0, 0.0])
b = torch.tensor([0.0, 1.0, 0.0, 1.0])
```

1. Concatenate (`torch.cat`) and print the length. Add them and print the length. Which one can you feed to a linear layer that still expects width 4?
2. L2-normalize `a` and `b`, then print cosine. Concat does not have a cosine until you define one; add already lives in the same $\mathbb{R}^4$.
3. Write four sentences: what a spectrogram frame is, why a hop that is too large smears 440 Hz versus 880 Hz in time, what a patch token threw away (phase, exact bin phase), and one fusion choice (concat vs add vs cross-attention) you would pick if the text vector were width 768 and the audio vector were width 512.